

# Alex Economopoulos

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## Skills

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**Software:** C++, Python, JavaScript, Java, HTML, CSS, MATLAB, Simulink, GUI Development, Git

**Hardware:** SolidWorks, AutoCAD, ANSYS, 3D Printing

**Design:** Debugging, Root Cause Analysis, Teamwork, Effective Communication

## Projects

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### Graphing Calculator (C++)

- Developed a real-time mathematical function plotting application with **custom function parsing**
- Implemented zooming and panning using **coordinate transformations**
- Designed interactive **GUI** using **OpenGL** and **ImGui**

### Sudoku Solver with GUI (Java)

- Designed and implemented a **recursive backtracking algorithm** to instantly solve 9x9 Sudoku puzzles
- Created a user-friendly **GUI** using **Swing GUI** for puzzle input and visualization

### 2D Collision Simulator (Python)

- Simulated elastic and inelastic collisions with user-controlled mass, velocity, and position
- Applied conservation laws to compute post-collision states and energy loss
- Developed an interactive **GUI** using **Tkinter**

## Experience

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### Guidance, Navigation, Controls Member

UW ORBITAL, Waterloo, ON

September 2025 – Present

- Simulated satellite eclipse periods using vector math and **Simulink** to predict solar panel charging capabilities
- Modelled expected Earth magnetic field measurements at the satellite using **MATLAB/Simulink** to support magnetorquer-based attitude stabilization
- Estimated Viscous and Coulomb friction coefficients to support **high fidelity reaction wheel model**

### Engineering Intern

Hammerschlag and Joffe Engineering Consultancy, Toronto, ON

July – August 2024

- Updated fire protection layouts by mapping legacy devices for a large department store chain using **AutoCAD**
- Performed **electrical load and water usage calculations** for corporate clients to aid engineering analyses
- Reviewed tender documents and contractor submissions, ensuring accuracy and compliance with **Excel**
- Drafted and edited **shop drawing review reports** to support construction quality control

### Head of International Health Competition Club

FUTURE HEALTH PROFESSIONALS, Toronto, ON

September 2024 – June 2025

- Managed registration, finances, and event logistics for **60+** chapter members
- Organized and facilitated a **peer mentor program**, personally mentoring 6 students
- Produced and directed a **promotional video** that increased chapter membership by **25%**
- Competed in the Medical Innovation event, designed a **first responder drone concept** with a team

## Education

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### University of Waterloo, ON

September 2025 – June 2030

Candidate for Bachelor of Applied Science in Biomedical Engineering

GPA: 3.9

Relevant Courses:

Digital Computation, Data Structures and Algorithms, Matrices and Linear Systems, Human Factors in Design